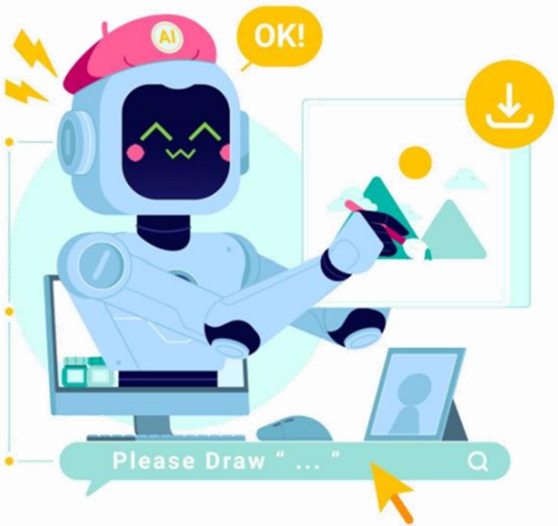
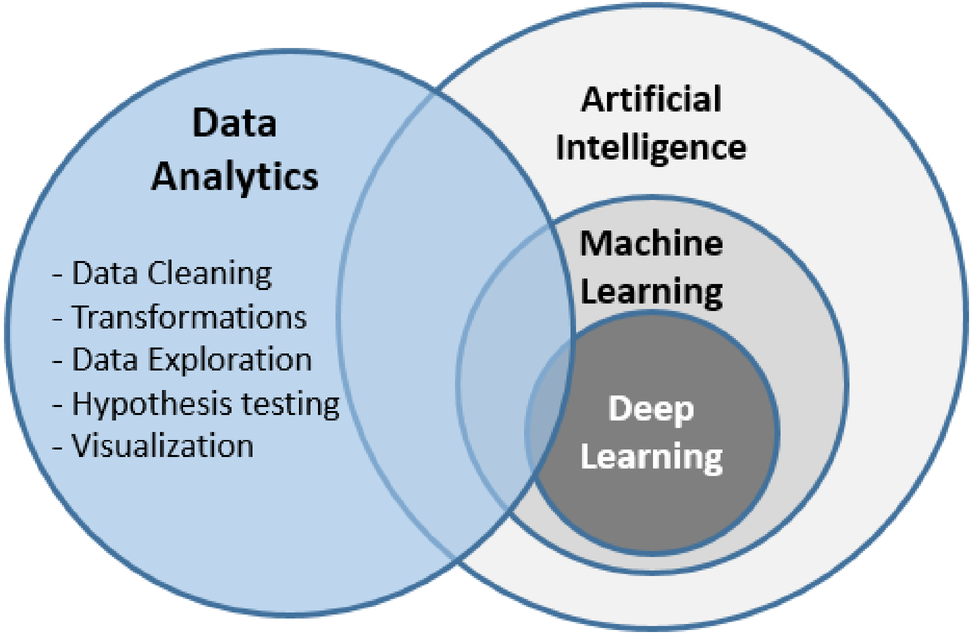
Introduction to Machine Learning

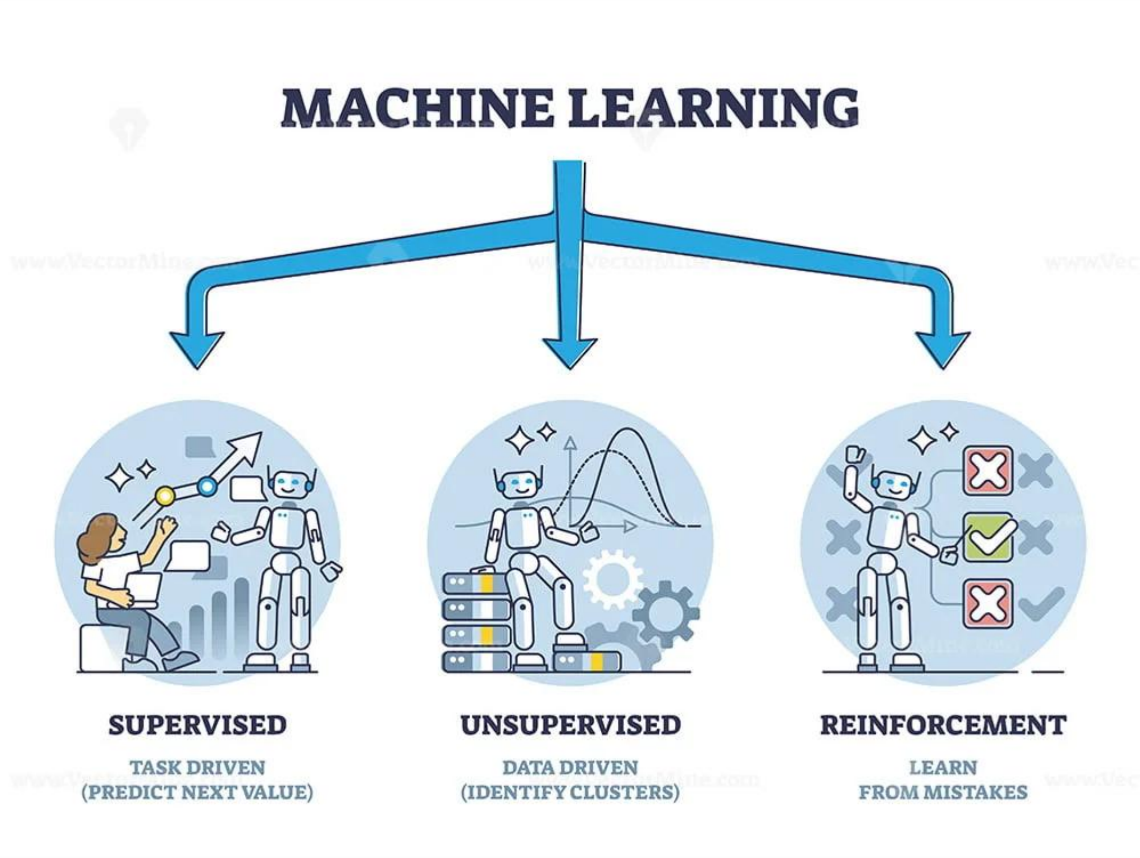



Data Analytics
- Data Cleaning
- Transformations
- Data Exploration
- Hypothesis testing
- Visualization
Artificial Intelligence
Machine Learning
Deep Learning

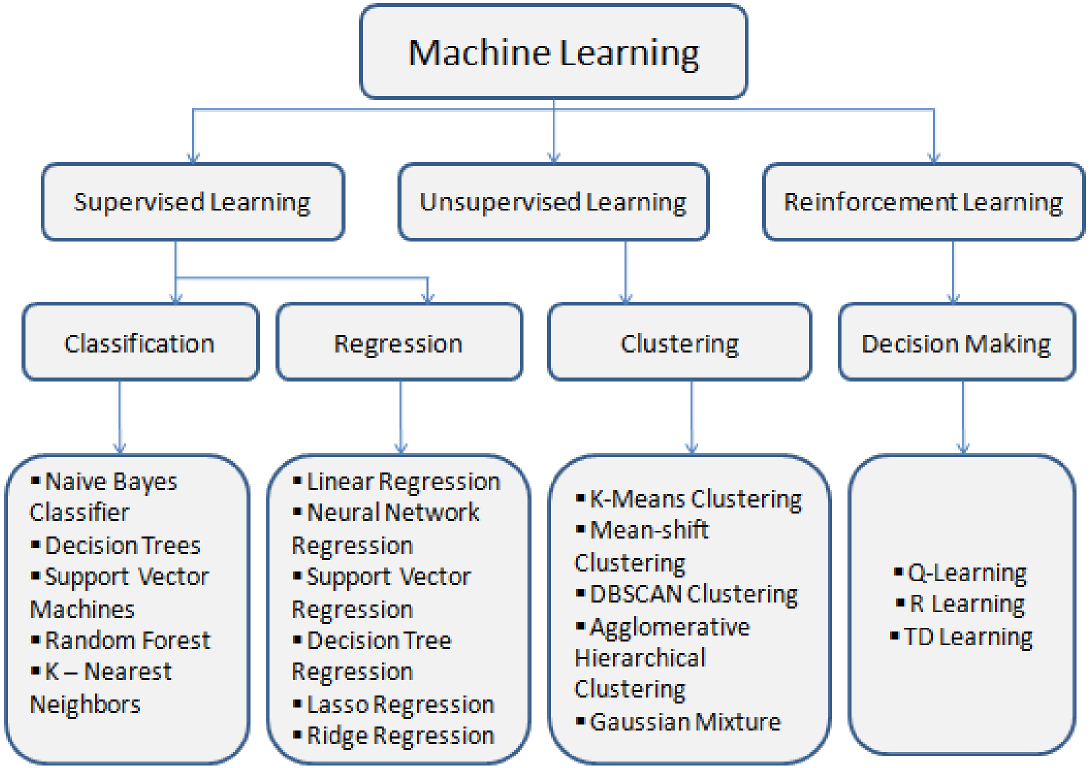
Machine Learning - การเรียนรู้ของเครื่อง

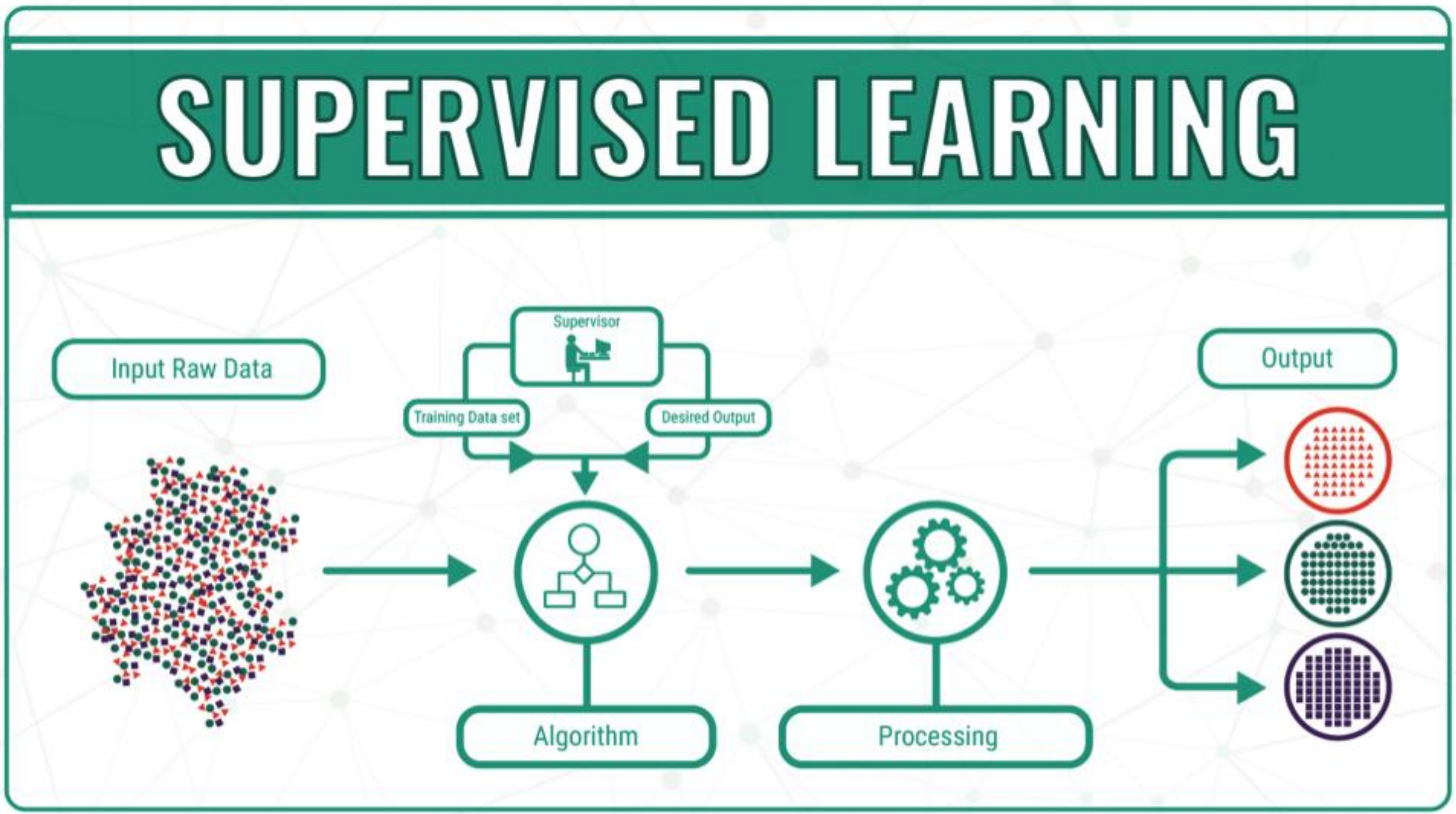
- ส่วนหนึ่งของปัญญาประดิษฐ์ (AI) ที่ทำให้ระบบคอมพิวเตอร์สามารถเรียนรู้จากข้อมูลได้โดยอัตโนมัติ โดยไม่ต้องเขียนโปรแกรมให้บอกว่าควรทำอย่างไรในทุกขั้นตอน
- กระบวนการที่ทำให้คอมพิวเตอร์สามารถรับรู้ข้อมูล ฝึกฝนตัวเอง และปรับปรุงความแม่นยำของผลลัพธ์ผ่านการเรียนรู้จากประสบการณ์

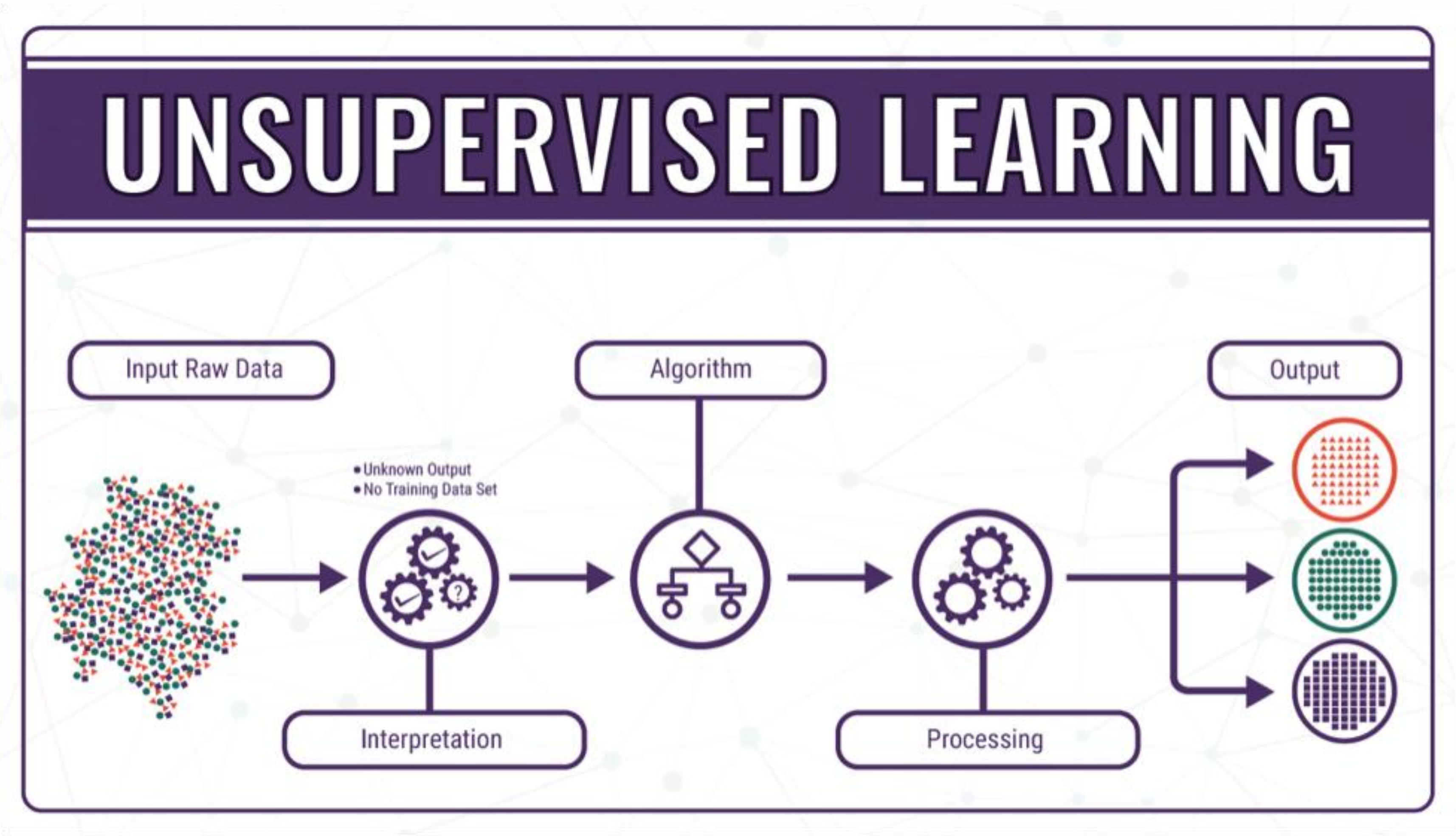
Types of Machine Learning

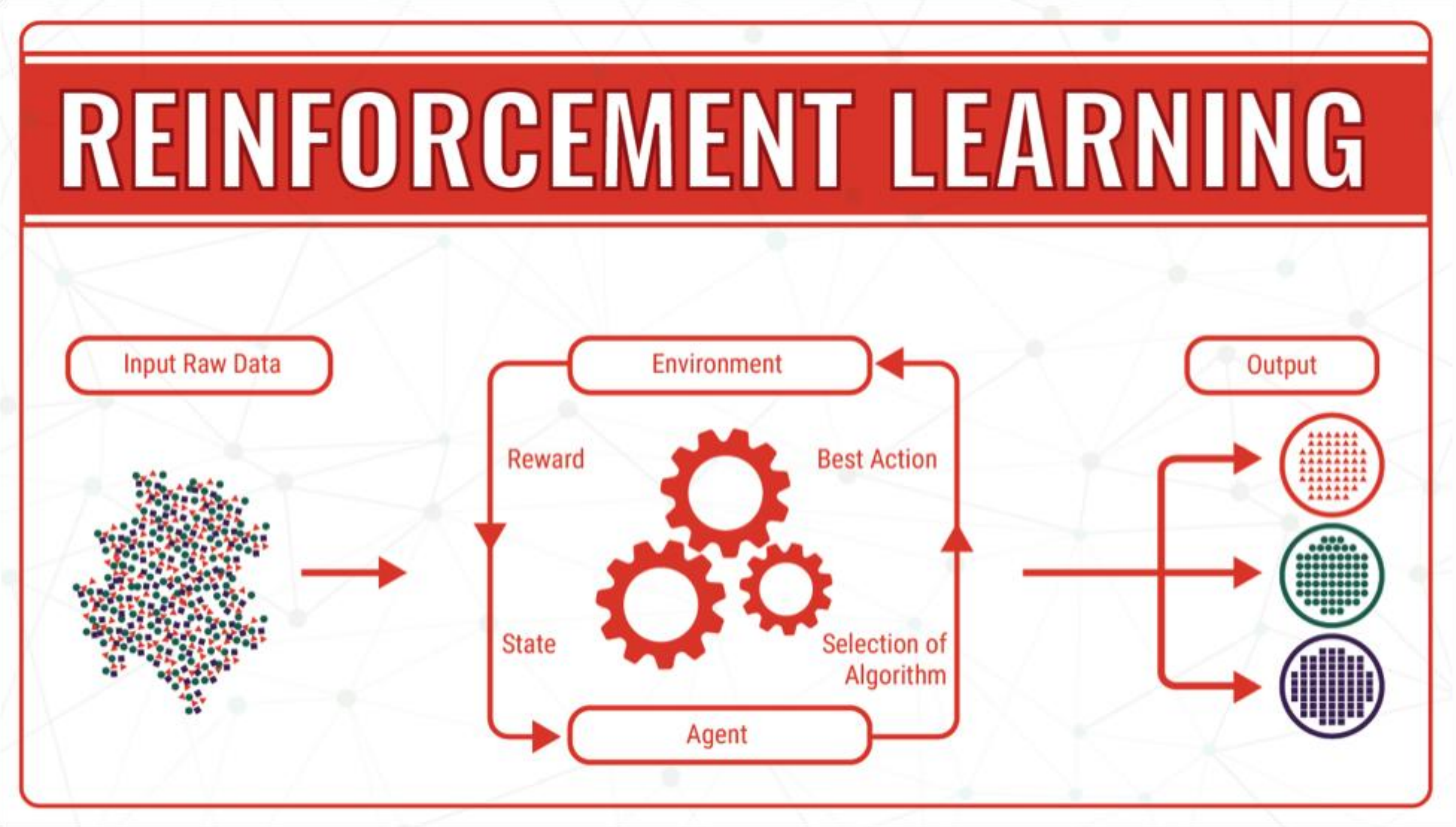


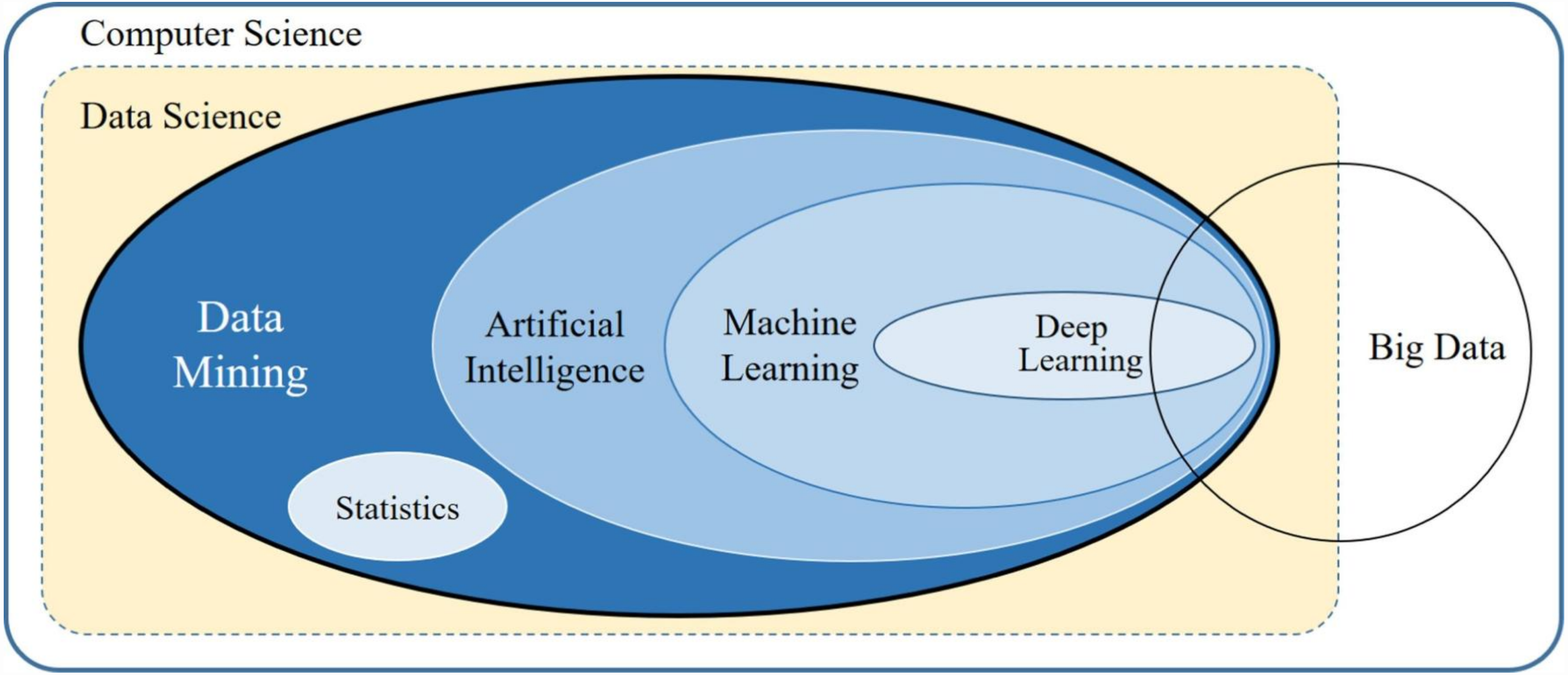
Types of Machine Learning




SUPERVISED LEARNING
Input Raw Data
Supervisor
Training Data set
Desired Output
Algorithm
Processing
Output


UNSUPERVISED LEARNING
Input Raw Data
Algorithm
Output
• Unknown Output
• No Training Data Set
Interpretation
Processing


REINFORCEMENT LEARNING
Input Raw Data
Environment
Output
Reward
Best Action
State
Selection of Algorithm
Agent



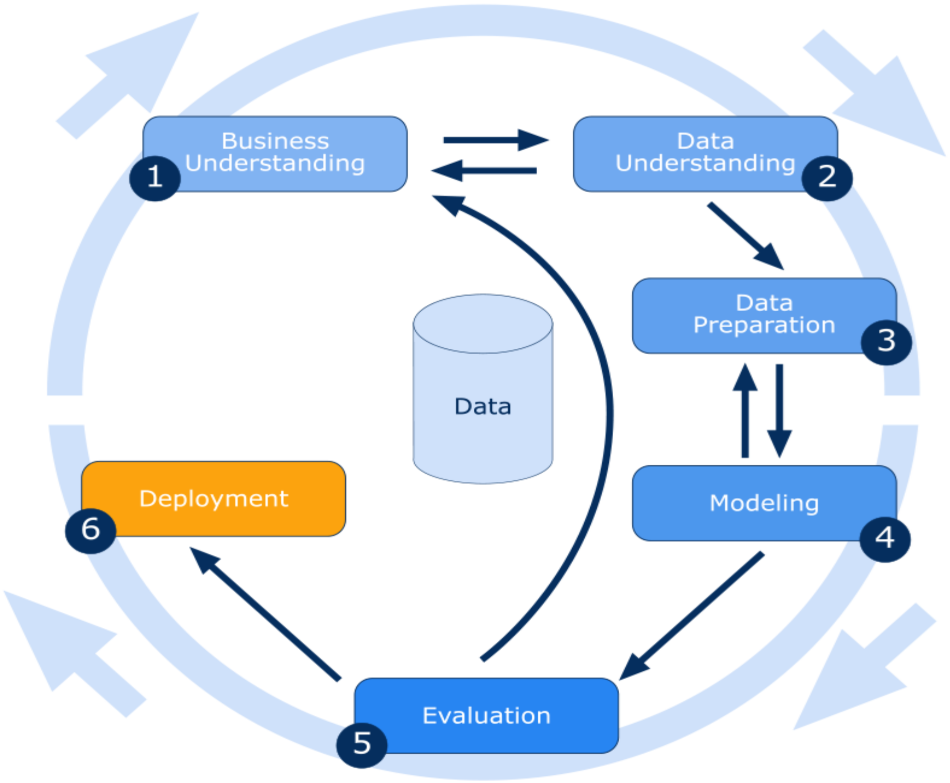
https://www.sciencedirect.com/topics/physics-and-astronomy/data-mining

Data Mining - การทำเหมืองข้อมูล

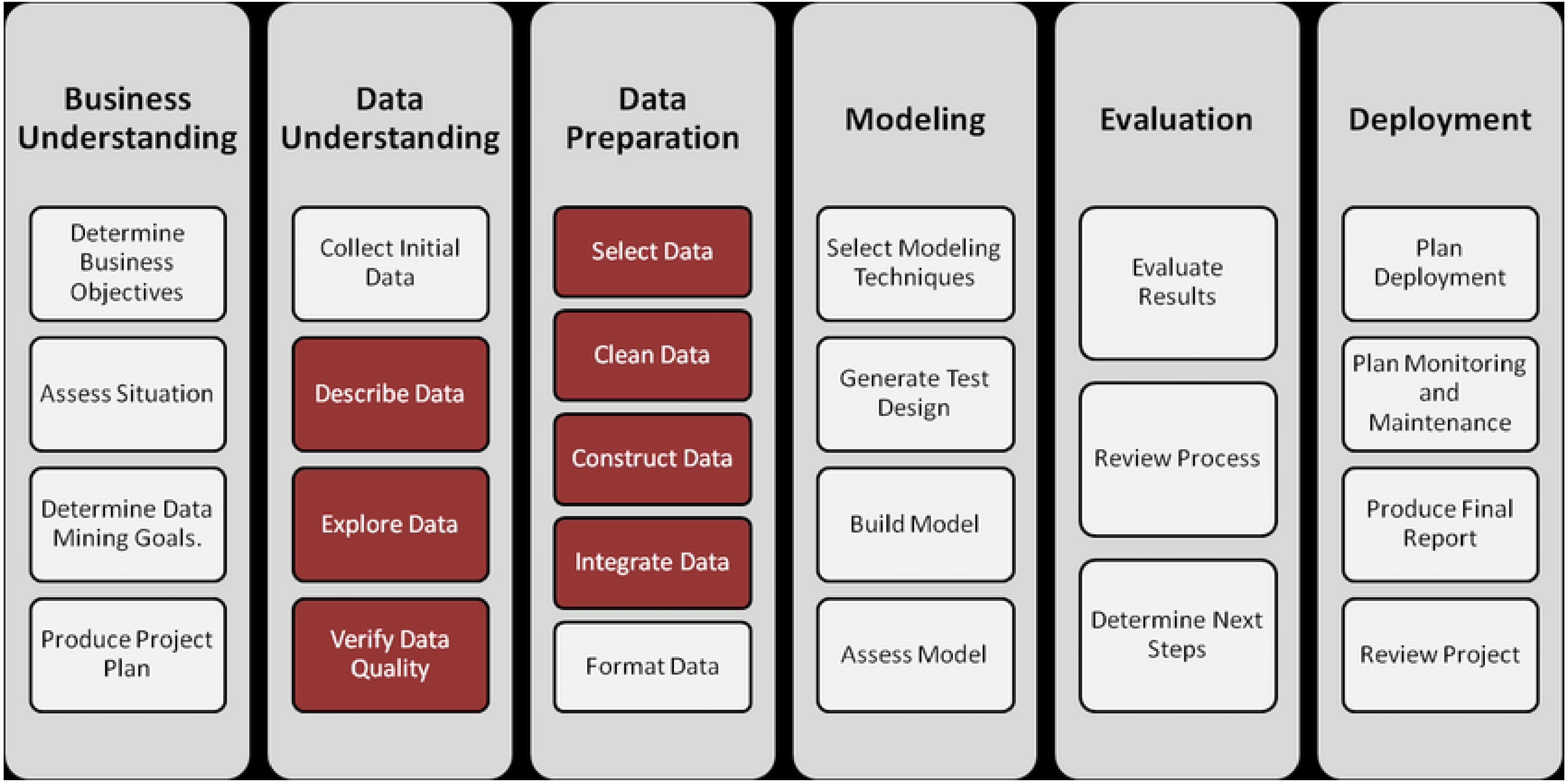
การวิเคราะห์ข้อมูลจากข้อมูลจำนวนมากเพื่อหาความสัมพันธ์ของข้อมูล

โดยทำการจำแนกประเภท แนวทาง และความสัมพันธ์ที่ซ่อนอยู่ในชุดข้อมูลนั้น

CRISP-DM
(Cross-Industry Standard Process for Data Mining)



CRISP-DM



Corrales, David & Ledezma Espino, Agapito & Corrales, Juan. (2015). A Conceptual Framework for Data Quality in Knowledge Discovery Tasks (FDQ-KDT): A Proposal. Journal of Computers. 10. 396-405. 10.17706/jcp.10.6.396-405.

01
K-means Clustering
การจัดกลุ่มด้วยขั้นตอนวิธี K-Means

Clustering

- การจัดกลุ่มข้อมูลตามความคล้ายคลึง (Similarity)
- จัดกลุ่มกันตามความแตกต่างของข้อมูล (Dissimilarity or Distance)
- Cluster Analysis
  - เป็นกระบวนการจัดข้อมูล/วัตถุต่างๆ ให้อยู่กลุ่มที่เหมาะสม โดยข้อมูล/วัตถุที่อยู่ในกลุ่มเดียวกันจะมีลักษณะคล้ายกัน แต่มีความแตกต่างจากข้อมูล/วัตถุในกลุ่มอื่น
- ไม่มีการกำหนดคลาสประเภทข้อมูลไว้ก่อนหรือไม่ทราบจำนวนกลุ่มล่วงหน้า เป็นการเรียนรู้แบบไม่มีผู้สอน (unsupervised learning)

Clustering Algorithms

K-Means:

A centroid-based algorithm that partitions data into k clusters, minimizing the variance within each cluster. It works by iteratively assigning data points to the nearest centroid and updating the centroid's position.

DBSCAN (Density-Based Spatial Clustering of Applications with Noise):

Groups data points based on density, identifying clusters as dense regions separated by sparser areas.

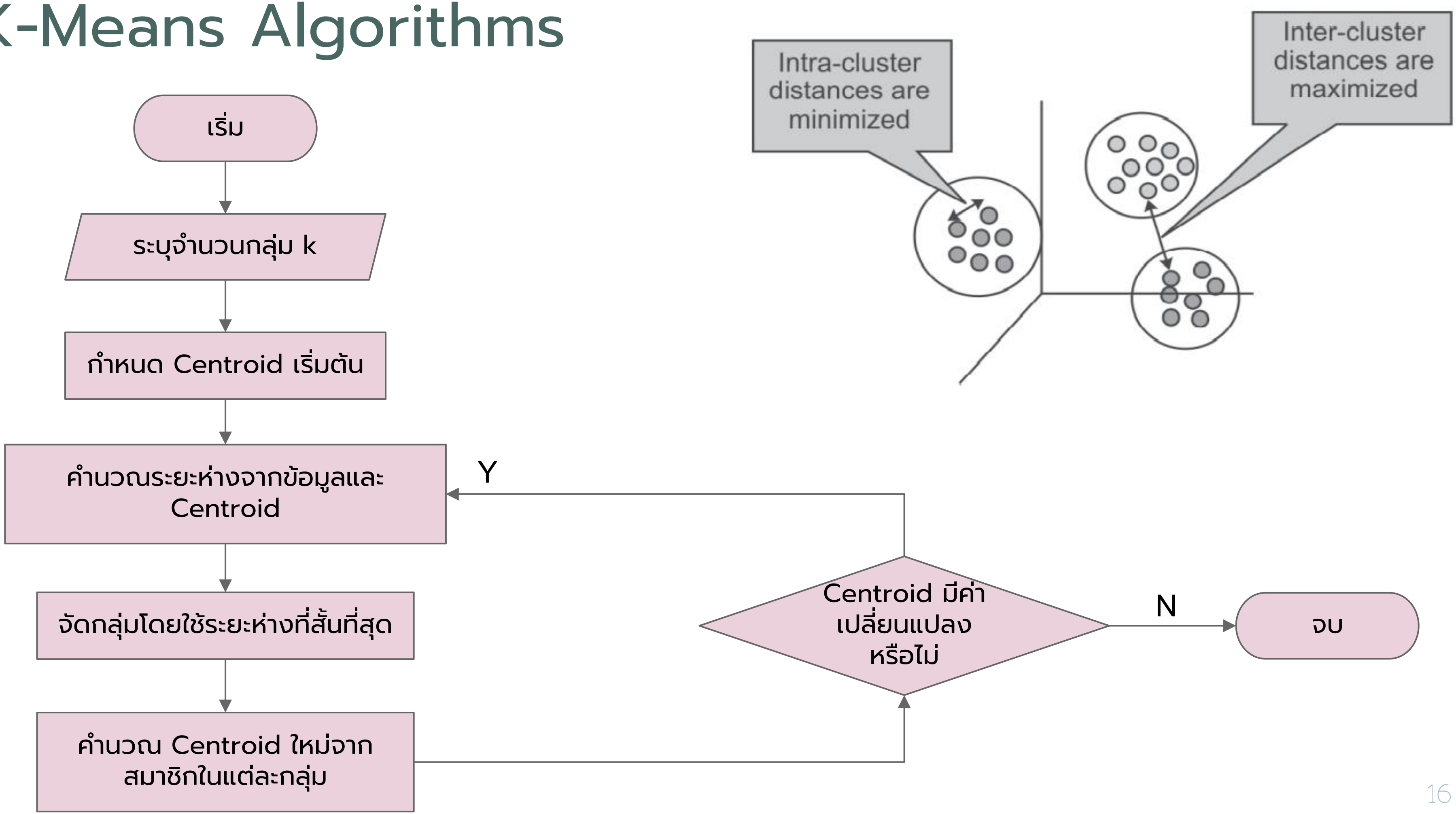
Hierarchical Clustering:

Builds a hierarchy of clusters, either by merging smaller clusters (agglomerative) or by splitting larger clusters (divisive). It can be visualized using a dendrogram.

Agglomerative Clustering:

A hierarchical clustering method that starts with each data point as its own cluster and merges the closest clusters iteratively.

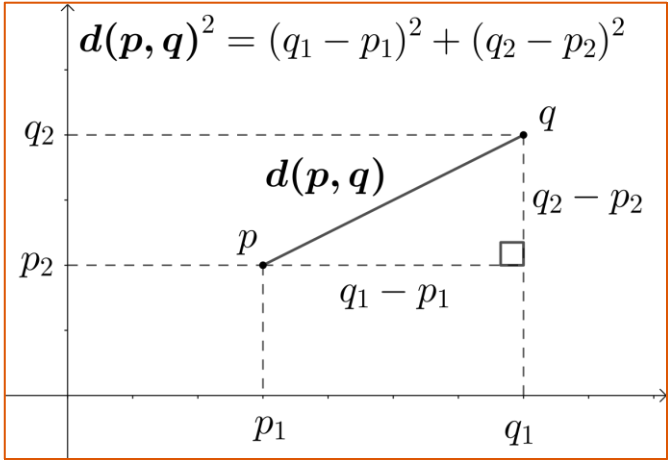
K-Means Algorithms

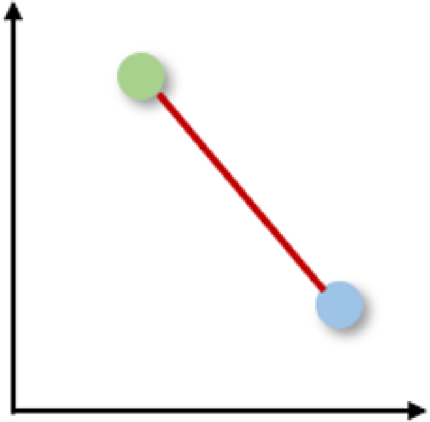


Euclidean distance

- โดยทั่วไปแล้ว k-means คำนวณระยะห่างระหว่างข้อมูลในแต่ละแถวกับ centroid ด้วย Euclidean distance

$$d(p,q) = d(q,p) = \sqrt{(q_1 - p_1)^2 + (q_2 - p_2)^2 + \cdots + (q_n - p_n)^2} = \sqrt{\sum_{i=1}^{n} (q_i - p_i)^2}$$





K-means Clustering Example: Customer Segmentation

Customer ID	Age (years)	Engagement with the page (days/weeks)
001	42	7
002	18	3
003	23	2
004	49	1
005	37	7
006	51	1
007	40	6
008	20	4

รอบที่ 1

1. ระบุจำนวนกลุ่ม k ค่าเริ่มต้น และกำหนด/สุ่ม Centroid เริ่มต้น

สมมติให้ k = 3

Customer ID	Age (years)	Engagement with the page (days/weeks)
C1 **001**	**42**	**7**
002	18	3
003	23	2
C2 **004**	**49**	**1**
005	37	7
C3 **006**	**51**	**1**
007	40	6
008	20	4

รอบที่ 1

2. หาค่าระยะห่างระหว่างข้อมูลกับจุด centroid ด้วยวิธี Euclidean Distance

ID	Age	Engagement
001	42	7
002	18	3
003	23	2
004	49	1
005	37	7
006	51	1
007	40	6
008	20	4

Distance from centroid C1

$$d(C1,2) = \sqrt{(42-18)^2 + (7-3)^2} = 24.33$$

$$d(C1,3) = \sqrt{(42- \quad)^2 + (7- \quad)^2} =$$

$$d(C1,4) = \sqrt{(42- \quad)^2 + (7- \quad)^2} =$$

$$d(C1,5) = \sqrt{(42- \quad)^2 + (7- \quad)^2} =$$

$$d(C1,6) = \sqrt{(42- \quad)^2 + (7- \quad)^2} =$$

$$d(C1,7) = \sqrt{(42- \quad)^2 + (7- \quad)^2} =$$

$$d(C1,8) = \sqrt{(42- \quad)^2 + (7- \quad)^2} =$$

รอบที่ 1

2. หาค่าระยะห่างระหว่างข้อมูลกับจุด centroid ด้วยวิธี Euclidean Distance

ID	Age	Engagement
001	42	7
002	18	3
003	23	2
004	49	1
005	37	7
006	51	1
007	40	6
008	20	4

Distance from centroid C2

$$d(C2,1) = \sqrt{(49-42)^2 + (1-7)^2} = 9.22$$

$$d(C2,2) = \sqrt{(49- \quad)^2 + (1- \quad)^2} =$$

$$d(C2,3) = \sqrt{(49- \quad)^2 + (1- \quad)^2} =$$

$$d(C2,5) = \sqrt{(49- \quad)^2 + (1- \quad)^2} =$$

$$d(C2,6) = \sqrt{(49- \quad)^2 + (1- \quad)^2} =$$

$$d(C2,7) = \sqrt{(49- \quad)^2 + (1- \quad)^2} =$$

$$d(C2,8) = \sqrt{(49- \quad)^2 + (1- \quad)^2} =$$

รอบที่ 1

2. หาค่าระยะห่างระหว่างข้อมูลกับจุด centroid ด้วยวิธี Euclidean Distance

ID	Age	Engagement
001	42	7
002	18	3
003	23	2
004	49	1
005	37	7
006	51	1
007	40	6
008	20	4

Distance from centroid C3

$$d(C3,1) = \sqrt{(51-42)^2 + (1-7)^2} = 10.82$$

$$d(C3,2) = \sqrt{(51- \quad)^2 + (1- \quad)^2} =$$

$$d(C3,3) = \sqrt{(51- \quad)^2 + (1- \quad)^2} =$$

$$d(C3,4) = \sqrt{(51- \quad)^2 + (1- \quad)^2} =$$

$$d(C3,5) = \sqrt{(51- \quad)^2 + (1- \quad)^2} =$$

$$d(C3,7) = \sqrt{(51- \quad)^2 + (1- \quad)^2} =$$

$$d(C3,8) = \sqrt{(51- \quad)^2 + (1- \quad)^2} =$$

รอบที่ 1

3. จัดกลุ่มโดยใช้ระยะห่างที่สั้นที่สุด

ID	Age	Engagement	Distance to C1	Distance to C2	Distance to C3	Cluster
001	**42**	**7**	0	9.22	10.82	
002	18	3	24.33	31.06	33.06	
003	23	2	19.65	26.02	28.02	
004	**49**	**1**	9.22	0	2.00	
005	37	7	5.00	13.42	15.23	
006	**51**	**1**	10.82	28.02	0	
007	40	6	2.24	10.30	12.08	
008	20	4	22.20	29.15	31.15	

รอบที่ 1

4. คำนวณ Centroid ใหม่จากสมาชิกในแต่ละกลุ่ม (หาค่าเฉลี่ยของกลุ่ม)

ID	Age	Engagement	Cluster
001	42	7	1
002	18	3	1
003	23	2	1
004	49	1	2
005	37	7	1
006	51	1	3
007	40	6	1
008	20	4	1

Cluster 1
- Mean of Age = 30.00
- Mean of Eng = 4.83

Cluster 2
- Mean of Age = 49.00
- Mean of Eng = 1.00

Cluster 3
- Mean of Age = 51.00
- Mean of Eng = 1.00

รอบที่ 2

2. หาค่าระยะห่างระหว่างข้อมูลกับจุด centroid ด้วยวิธี Euclidean Distance

centroid C1 = (30, 4.83) centroid C2 = (49, 1) centroid C3 = (51, 1)

ID	Age	Engagement
001	42	7
002	18	3
003	23	2
004	49	1
005	37	7
006	51	1
007	40	6
008	20	4

$$d(C1,1) = \sqrt{(30-42)^2 + (4.83-7)^2} = 12.20$$

$$d(C1,2) = \sqrt{(30- \quad)^2 + (4.83- \quad)^2} =$$

$$d(C1,3) = \sqrt{(30- \quad)^2 + (4.83- \quad)^2} =$$

$$d(C1,4) = \sqrt{(30- \quad)^2 + (4.83- \quad)^2} =$$

$$d(C1,5) = \sqrt{(30- \quad)^2 + (4.83- \quad)^2} =$$

$$d(C1,6) = \sqrt{(30- \quad)^2 + (4.83- \quad)^2} =$$

$$d(C1,7) = \sqrt{(30- \quad)^2 + (4.83- \quad)^2} =$$

$$d(C1,8) = \sqrt{(30- \quad)^2 + (4.83- \quad)^2} =$$

รอบที่ 2

หาค่าระยะห่างระหว่างข้อมูลกับจุด centroid C2 และ centroid C3

รอบที่ 2

3. จัดกลุ่มโดยใช้ระยะห่างที่สั้นที่สุด

ID	Age	Engagement	Distance to C1	Distance to C2	Distance to C3	Cluster
001	42	7	12.20	9.22	10.82	
002	18	3	12.14	31.06	33.06	
003	23	2	7.55	26.02	28.02	
004	49	1	19.38	0	2.00	
005	37	7	7.33	13.42	15.23	
006	51	1	21.35	28.02	0	
007	40	6	10.07	10.30	12.08	
008	20	4	10.03	29.15	31.15	

รอบที่ 2

4. คำนวณ Centroid ใหม่จากสมาชิกในแต่ละกลุ่ม (หาค่าเฉลี่ยของกลุ่ม)

ID	Age	Engagement	Cluster
001	42	7	2
002	18	3	1
003	23	2	1
004	49	1	2
005	37	7	1
006	51	1	3
007	40	6	1
008	20	4	1

Cluster 1
- Mean of Age = 27.6
- Mean of Eng = 4.4

Cluster 2
- Mean of Age = 45.5
- Mean of Eng = 4.00

Cluster 3
- Mean of Age = 51.00
- Mean of Eng = 1.00

รอบที่ 3

2. หาค่าระยะห่างระหว่างข้อมูลกับจุด centroid ด้วยวิธี Euclidean Distance

centroid C1 = (27.6, 4.4)   centroid C2 = (45.5, 4)   centroid C3 = (51, 1)

ID	Distance to C1	Distance to C2	Distance to C3	Cluster
001	14.62	4.61	10.82	2
002	9.70	27.52	33.06	1
003	5.19	22.59	28.02	1
004	21.67	4.61	2.00	3
005	9.75	9.01	15.23	2
006	23.65	6.27	0	3
007	12.50	5.85	12.08	2
008	7.61	25.5	31.14	1

รอบที่ 3

4. คำนวณ Centroid ใหม่จากสมาชิกในแต่ละกลุ่ม (หาค่าเฉลี่ยของกลุ่ม)

ID	Age	Engagement	Cluster
001	42	7	2
002	18	3	1
003	23	2	1
004	49	1	3
005	37	7	2
006	51	1	3
007	40	6	2
008	20	4	1

Cluster 1
- Mean of Age = 20.33
- Mean of Eng = 3.0

Cluster 2
- Mean of Age = 39.67
- Mean of Eng = 6.67

Cluster 3
- Mean of Age = 50.0
- Mean of Eng = 1.0

รอบที่ 4

2. หาค่าระยะห่างระหว่างข้อมูลกับจุด centroid ด้วยวิธี Euclidean Distance

centroid C1 = (20.33, 3.0)   centroid C2 = (39.67, 6.67)   centroid C3 = (50.0, 1.0)

ID	Distance to C1	Distance to C2	Distance to C3	Cluster
001	22.03	2.36	10.0	2
002	2.33	21.97	32.06	1
003	2.85	17.31	27.02	1
004	28.74	10.92	1.0	3
005	17.14	2.69	14.32	2
006	30.73	12.67	1.0	3
007	19.89	0.75	11.18	2
008	1.05	19.85	30.15	1

รอบที่ 4

4. คำนวณ Centroid ใหม่จากสมาชิกในแต่ละกลุ่ม (หาค่าเฉลี่ยของกลุ่ม)

ID	Age	Engagement	Cluster
001	42	7	2
002	18	3	1
003	23	2	1
004	49	1	3
005	37	7	2
006	51	1	3
007	40	6	2
008	20	4	1

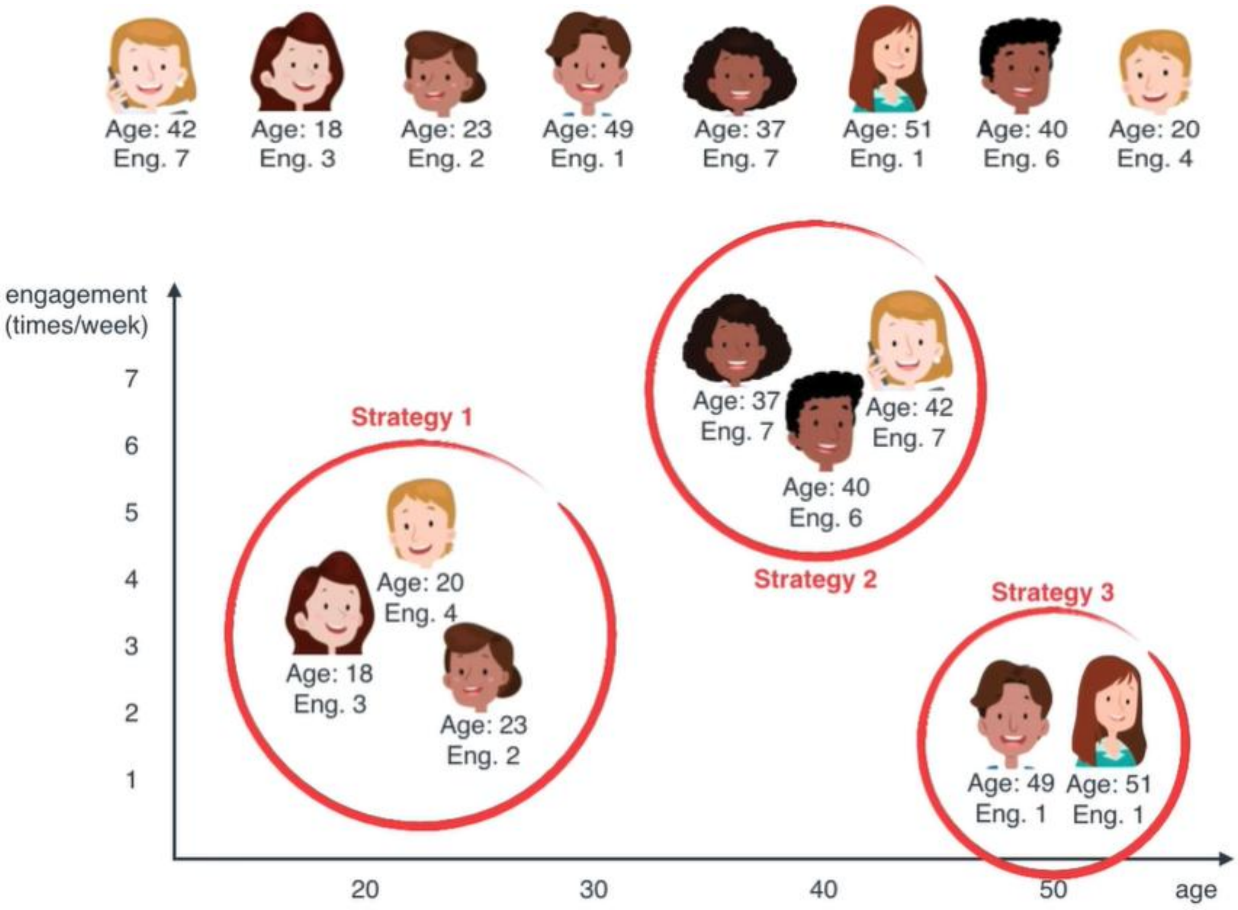
Cluster 1
- Mean of Age = 20.33
- Mean of Eng = 3.0

Cluster 2
- Mean of Age = 39.67
- Mean of Eng = 6.67

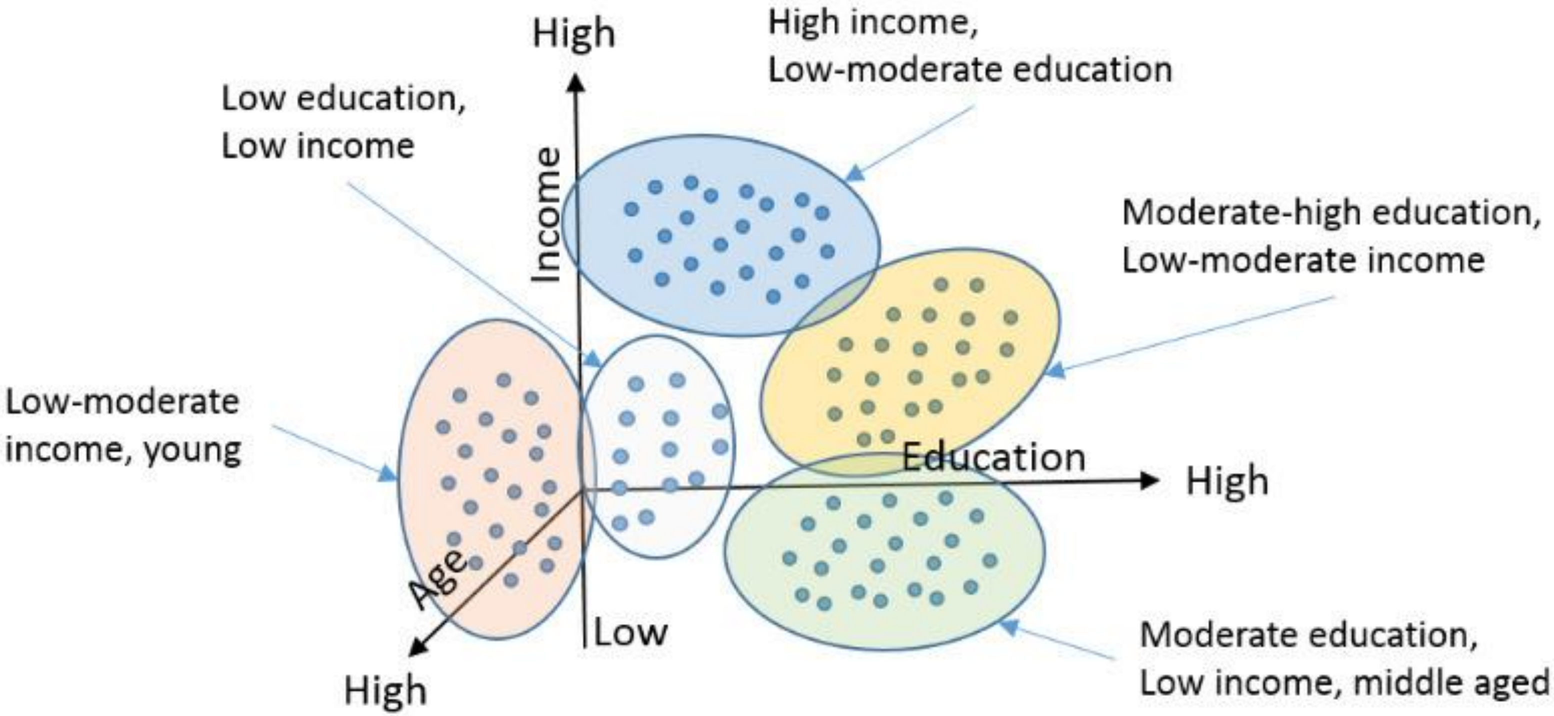
Cluster 3
- Mean of Age = 50.0
- Mean of Eng = 1.0

ได้ผลลัพธ์เหมือนรอบก่อนหน้านี้
แสดงว่า centroid ไม่เปลี่ยนแล้ว
จบการทำงาน

K-means Clustering Example: Customer Segmentation



Clustering - Example



https://www.researchgate.net/figure/Example-3-of-K-means-clustering-using-R_fig1_313066371

Performance Evaluation

- Within Cluster Sum of Squares (WCSS)
- Silhouette coefficient score
- Davies-Bouldin Index
- Calinski-Harabasz Index

Performance Evaluation: Within Cluster Sum of Squares (WCSS)

- also known as inertia
- measures the compactness of clusters in a dataset by calculating the sum of squared distances between each data point and its cluster's centroid.
- lower WCSS values indicate tighter, more cohesive clusters, while higher values suggest less compact clusters.

$$\text{WCSS} = \sum_{C_k}^{C_n}\Big(\sum_{d_i \, in \, C_i}^{d_m} distance(d_i, C_k)^2\Big)$$

Where,

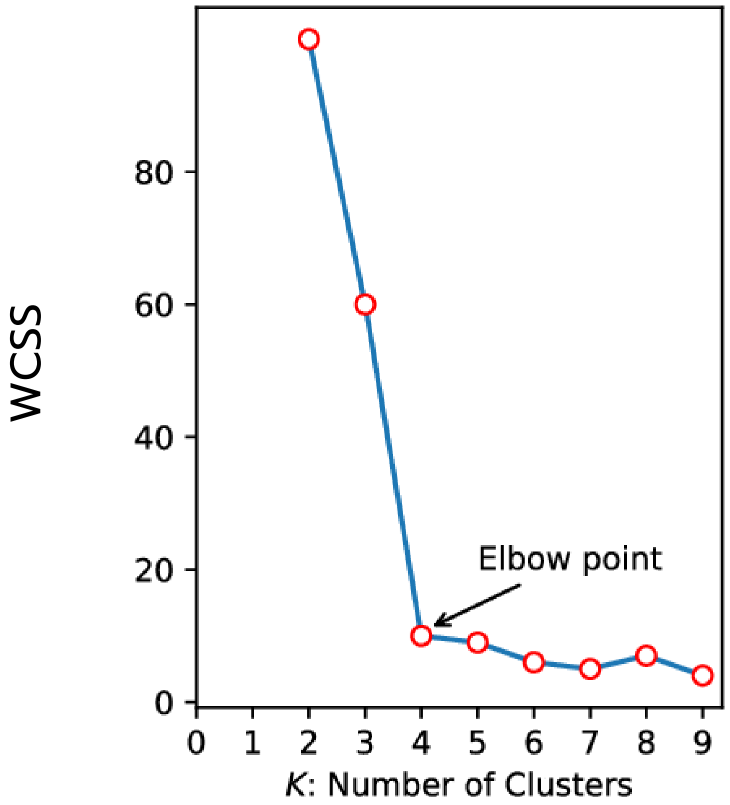
C is the cluster centroids and d is the data point in each Cluster.

คำนวณ Within Cluster Sum of Squares จากตัวอย่าง

centroid C1 = (20.33, 3.0)   centroid C2 = (39.67, 6.67)   centroid C3 = (50.0, 1.0)

ID	Distance to C1	Distance to C2	Distance to C3	Cluster
001		2.36		2
002	2.33			1
003	2.85			1
004			1.0	3
005		2.69		2
006			1.0	3
007		0.75		2
008	1.05			1

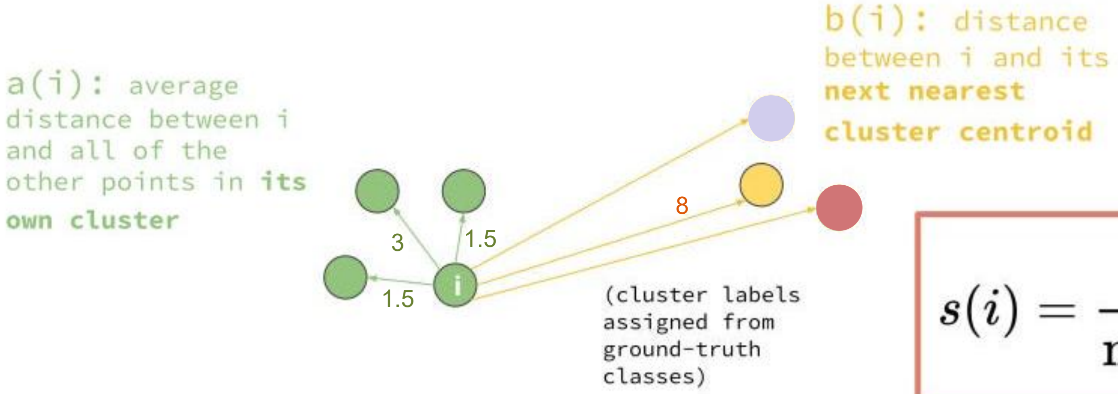
Optimal Number of Clusters



Performance Evaluation: Silhouette coefficient score

- It measures how similar an object is to its own cluster (cohesion) compared to other clusters (separation).
- The score ranges from -1 to +1, with higher values indicating better clustering.
  - +1: data point is well-clustered, far away from other clusters.
  - 0: data point lies on or very close to the decision boundary between two clusters.
  - -1: data point may have been assigned to the wrong cluster.

The overall Silhouette Score for the dataset is the average of the Silhouette Scores for all data points.



For a single point, i

$$\boxed{s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}}}$$

Clustering performance evaluation

Silhouette Coefficient:

- Measures how similar a point is to its own cluster compared to other clusters.
- Range: -1 to 1 → Higher is better

Advantages:

- The score is bounded between -1 for incorrect clustering and +1 for highly dense clustering. Scores around zero indicate overlapping clusters.
- The score is higher when clusters are dense and well separated, which relates to a standard concept of a cluster.

Drawbacks:

- The Silhouette Coefficient is generally higher for convex clusters than other concepts of clusters, such as density based clusters like those obtained through DBSCAN.

https://scikit-learn.org/stable/modules/clustering.html#clustering-evaluation

Clustering performance evaluation

Davies-Bouldin Index:

- This index signifies the average 'similarity' between clusters, where the similarity is a measure that compares the distance between clusters with the size of the clusters themselves.
- Lower Davies-Bouldin scores indicate better clustering, with values closer to zero suggesting well-separated and compact clusters.

Clustering performance evaluation

Davies-Bouldin Index:

Mathematically, if we have k clusters, the DBI is often defined as:

$$\text{DBI} = \frac{1}{k}\sum_{i=1}^{k} R_i \quad \text{where} \quad R_i = \max_{j \neq i}\left(\frac{s_i + s_j}{d(c_i, c_j)}\right)$$

where:

- s_i represents the dispersion (e.g., average distance) within cluster i
- $d(c_i, c_j)$ denotes the distance between the centroids c_i and c_j of clusters i and j, respectively.

A lower DBI indicates superior clustering performance, as it suggests that clusters are compact and well-separated.

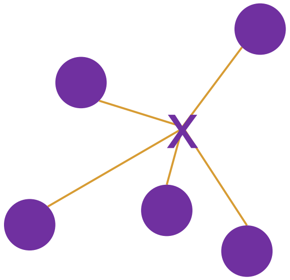
Clustering performance evaluation

Davies-Bouldin Index:

- s_i:
  This term represents the average distance between each point in cluster i and its centroid c_i. It can be computed using the formula:

$$s_i = \frac{1}{|C_i|} \sum_{x \in C_i} d(x, c_i)$$

  Here, C_i is the set of points in the i-th cluster, $|C_i|$ is the number of points in that cluster, and $d(x, c_i)$ is the distance metric (e.g., Euclidean) between the data point x and the centroid c_i.



- **Low Intra-Cluster Distances (s_i):**
  When the data points within a cluster are tightly bound around the centroid, s_i is small. This indicates high compactness.

Clustering performance evaluation

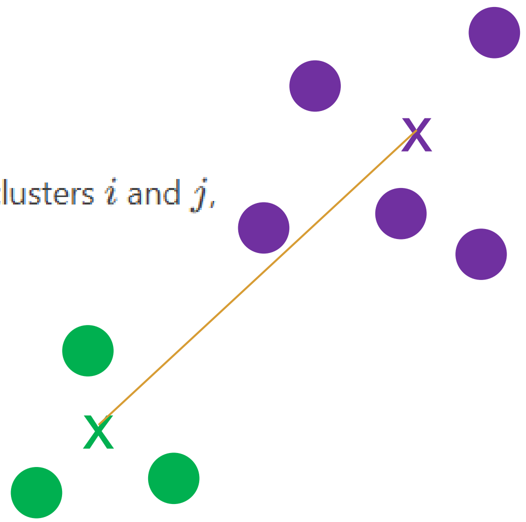
Davies-Bouldin Index:

- $d(c_i, c_j)$:
  This term calculates the distance between the centroids c_i and c_j of clusters i and j, respectively. For Euclidean space, it is computed as:

$$d(c_i, c_j) = \sqrt{\sum_{l=1}^{n} (c_{i,l} - c_{j,l})^2}$$

  where n is the number of dimensions in the data.

- **High Inter-Cluster Distances ($d(c_i, c_j)$):**
  Larger distances between centroids imply that clusters are well-separated.



Clustering performance evaluation

Davies-Bouldin Index:

$$\text{DBI} = \frac{1}{k}\sum_{i=1}^{k} R_i \quad \text{where} \quad R_i = \max_{j \neq i}\left(\frac{s_i + s_j}{d(c_i, c_j)}\right)$$

- **Low Intra-Cluster Distances (s_i):**

  When the data points within a cluster are tightly bound around the centroid, s_i is small. This indicates high compactness.

- **High Inter-Cluster Distances ($d(c_i, c_j)$):**

  Larger distances between centroids imply that clusters are well-separated.

Thus, a low DBI reflects the ideal scenario where s_i values are minimal relative to $d(c_i, c_j)$ values across all clusters.

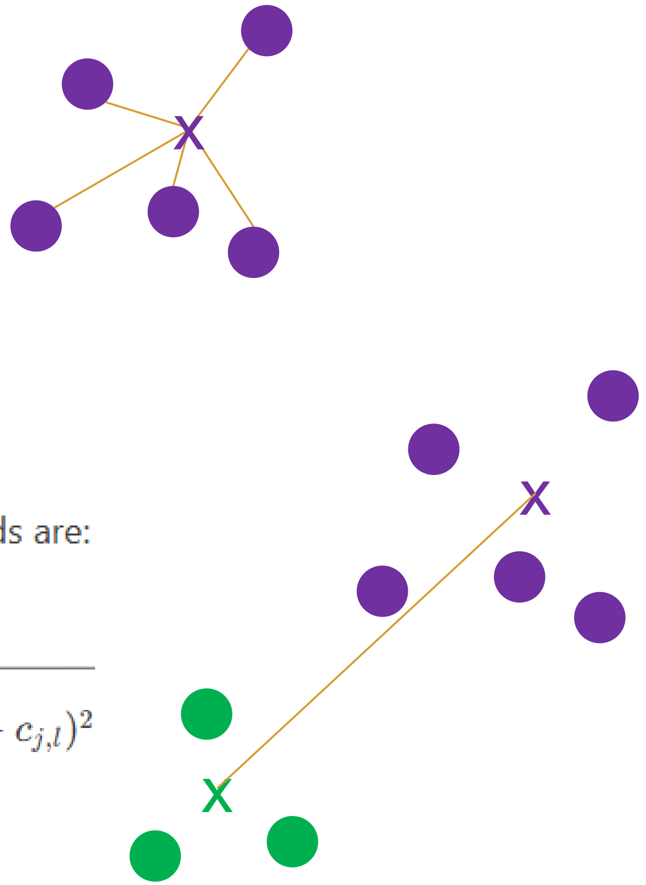
Clustering performance evaluation

Davies-Bouldin Index:

Consider a simple scenario with three clusters, where:

- Cluster 1: $s_1 = 2$
- Cluster 2: $s_2 = 3$
- Cluster 3: $s_3 = 2.5$

$$s_i = \frac{1}{|C_i|} \sum_{x \in C_i} d(x, c_i)$$



Assume the pairwise distances between cluster centroids are:

- $d(c_1, c_2) = 8$
- $d(c_1, c_3) = 7$
- $d(c_2, c_3) = 9$

$$d(c_i, c_j) = \sqrt{\sum_{l=1}^{n} (c_{i,l} - c_{j,l})^2}$$

Clustering performance evaluation

Davies-Bouldin Index:

$$\text{DBI} = \frac{1}{k}\sum_{i=1}^{k} R_i \quad \text{where} \quad \boxed{R_i = \max_{j \neq i}\left(\frac{s_i + s_j}{d(c_i, c_j)}\right)}$$

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Assume the pairwise distances between cluster centroids are:

- $d(c_1, c_2) = 8$
- $d(c_1, c_3) = 7$
- $d(c_2, c_3) = 9$

For each cluster, we compute the maximum ratio:

- **For Cluster 1:**

$$R_1 = \max\left(\frac{s_1 + s_2}{d(c_1, c_2)}, \frac{s_1 + s_3}{d(c_1, c_3)}\right)$$

$$= \max\left(\frac{2+3}{8}, \frac{2+2.5}{7}\right)$$

Calculations:

$\frac{5}{8} \approx 0.625$ and $\frac{4.5}{7} \approx 0.643$.

Hence, $R_1 \approx 0.643$.

Clustering performance evaluation

Davies-Bouldin Index:

Consider a simple scenario with three clusters, where:

- Cluster 1: $s_1 = 2$
- Cluster 2: $s_2 = 3$
- Cluster 3: $s_3 = 2.5$

Assume the pairwise distances between cluster centroids are:

- $d(c_1, c_2) = 8$
- $d(c_1, c_3) = 7$
- $d(c_2, c_3) = 9$

- **For Cluster 2:**

$$R_2 = \max\left(\frac{s_2 + s_1}{d(c_1, c_2)}, \frac{s_2 + s_3}{d(c_2, c_3)}\right)$$

$$= \max\left(\frac{3+2}{8}, \frac{3+2.5}{9}\right)$$

Calculations:

$\frac{5}{8} \approx 0.625$ and $\frac{5.5}{9} \approx 0.611.$

Hence, $R_2 \approx 0.625.$

Clustering performance evaluation

Davies-Bouldin Index:

Consider a simple scenario with three clusters, where:

- Cluster 1: $s_1 = 2$
- Cluster 2: $s_2 = 3$
- Cluster 3: $s_3 = 2.5$

Assume the pairwise distances between cluster centroids are:

- $d(c_1, c_2) = 8$
- $d(c_1, c_3) = 7$
- $d(c_2, c_3) = 9$

- **For Cluster 3:**

$$R_3 = \max\left(\frac{s_3 + s_1}{d(c_1, c_3)}, \frac{s_3 + s_2}{d(c_2, c_3)}\right):$$

$$= \max\left(\frac{2.5 + 2}{7}, \frac{2.5 + 3}{9}\right)$$

Calculations:

$\frac{4.5}{7} \approx 0.643$ and $\frac{5.5}{9} \approx 0.611$.

Hence, $R_3 \approx 0.643$.

Clustering performance evaluation

Davies-Bouldin Index:

Finally, the overall DBI is calculated as:

$$\text{DBI} = \frac{R_1 + R_2 + R_3}{3} \approx \frac{0.643 + 0.625 + 0.643}{3} \approx 0.637$$

This step-by-step example illustrates the methodical approach involved in calculating the DBI and highlights the balance between compactness and separation.

Clustering performance evaluation

Davies-Bouldin Index:

Advantages:

- The computation of Davies-Bouldin is simpler than that of Silhouette scores.
- The index is solely based on quantities and features inherent to the dataset as its computation only uses point-wise distances.

Drawbacks:

- The Davies-Bouldin index is generally higher for convex clusters than other concepts of clusters, such as density-based clusters like those obtained from DBSCAN.
- The usage of centroid distance limits the distance metric to Euclidean space.

Clustering performance evaluation

Calinski-Harabasz Index:

- also known as the Variance Ratio Criterion (VRC)
- It measures the ratio of between-cluster dispersion to within-cluster dispersion, indicating how well-separated and dense the clusters are
- Higher indicates better separation and compactness

$$CH = \frac{BCSS/(k-1)}{WCSS/(n-k)}$$

Clustering performance evaluation

Calinski-Harabasz Index:

$$CH = \frac{BCSS/(k-1)}{WCSS/(n-k)}$$

BCSS (Between-Cluster Sum of Squares) is the weighted sum of squared Euclidean distances between each cluster centroid (mean) and the overall data centroid (mean).

$$BCSS = \sum_{i=1}^{k} n_i ||\mathbf{c}_i - \mathbf{c}||^2$$

WCSS (Within-Cluster Sum of Squares) is the sum of squared Euclidean distances between the data points and their respective cluster centroids:

$$WCSS = \sum_{i=1}^{k} \sum_{\mathbf{x} \in C_i} ||\mathbf{x} - \mathbf{c}_i||^2$$

Clustering performance evaluation

Calinski-Harabasz Index:

จุด	X	Y	Cluster
A	1	2	1
B	2	1	1
C	4	5	2
D	5	4	2

$$BCSS = \sum_{i=1}^{k} n_i ||\mathbf{c}_i - \mathbf{c}||^2$$

Cluster 1: n=2

2 * [(1.5-3)^2 + [(1.5-3)^2] = 9.0

Cluster 2: n=2

2 * [(4.5-3)^2 + [(4.5-3)^2] = 9.0

BCSS = 9.0 + 9.0 = 18

BCSS (Between-Cluster Sum of Squares) is the weighted sum of squared Euclidean distances between each cluster centroid and the overall data centroid.

Cluster 1: centroid = (1.5, 1.5)
Cluster 2: centroid = (4.5, 4.5)

(3.0, 3.0)

- High BSS indicates that clusters are well separated.

Clustering performance evaluation

Calinski-Harabasz Index:

จุด	X	Y	Cluster
A	1	2	1
B	2	1	1
C	4	5	2
D	5	4	2

$$WCSS = \sum_{i=1}^{k} \sum_{\mathbf{x} \in C_i} ||\mathbf{x} - \mathbf{c}_i||^2$$

Cluster 1:

d(A,centroid1) = (1.5-1)^2 + [(1.5-2)^2 = 0.5

d(B,centroid1) = (1.5-2)^2 + [(1.5-1)^2 = 0.5

รวม 1.0

Cluster 2:

d(C,centroid1) = (4.5-4)^2 + [(4.5-5)^2 = 0.5

d(C,centroid1) = (4.5-5)^2 + [(4.5-4)^2 = 0.5

รวม 1.0

WCSS = 1.0 + 1.0 = 2

WCSS (Within-Cluster Sum of Squares) is the sum of squared Euclidean distances between the data points and their respective cluster centroids.

- Low WSS signifies that data points within clusters are closely grouped.

Clustering performance evaluation

Calinski-Harabasz Index:

$$CH = \frac{\overbrace{BCSS}^{18}/(\overbrace{k}^{2}-1)}{\underbrace{WCSS}_{2}/(\underbrace{n}_{4}-\underbrace{k}_{2})} = 18$$

- High CH Index Value: Suggests that the clusters are well defined—high between-cluster separation combined with low within-cluster dispersion.
- Low CH Index Value: Implies that the clustering structure may not be optimal. It could be due to clusters overlapping or being too spread out internally.

Clustering performance evaluation

Calinski-Harabasz Index:

Advantages:

- The score is higher when clusters are dense and well separated, which relates to a standard concept of a cluster.
- The score is fast to compute.

Drawbacks:

- The Calinski-Harabasz index is generally higher for convex clusters than other concepts of clusters, such as density based clusters like those obtained through DBSCAN.

แบบฝึกหัด

กระทรวงสาธารณสุขมีข้อมูลยาอยู่ 6 ชนิด ซึ่งมีค่า pH-Index และ Weight ดังแสดงในตาราง จงใช้วิธี K-means clustering เพื่อแบ่งยาออกเป็น 3 กลุ่ม

Medicine	pH-Index	Weight
a	1	4
b	2	3
d	3	4
d	5	1
e	6	2
f	4	1

แบบฝึกหัด LAB

ชื่อธัญพืช	น้ำตาล	แคลลอรี	โปรตีน	ไขมัน	โซเดียม
0-A- รำข้าว 100%	6	70	4	1	130
1-B- รำข้าวธรรมชาติ 100%	8	120	3	5	15
2-C - รำข้าวทั้งหมด	5	70	4	1	260
3-D- รำข้าวทั้งหมดที่สกัดจากใยอาหาร	0	50	4	0	140
4-E- อัลมอนด์	8	110	2	2	200
5-F- แอปเปิลอบเชย	10	110	2	2	180
6-G- เหล้าที่ทำจากผลแอปเปิล	14	110	2	0	125
7-H- พื้นฐาน 4	8	130	3	2	210
8-I- รำข้าว Chex	6	90	2	1	200
9-J- รำข้าว Flakes	5	90	3	0	210
10-K- Cap ‘n’ Crunch	12	120	1	2	220
11-L- Cheerios	1	110	6	2	290
12-M- อบเชยทอดกรอบ	9	120	1	3	210
13-N- Cluster	7	110	3	2	140
14-O- โกโก้	13	110	1	1	180